

# Finite-time Fluctuations and Long-chain Gallavotti–Cohen Diagnostics in the NLS Chain

Numerical validation report

30 August 2026

## Abstract

We report the completed hierarchy of fluctuation tests for the boundary-driven nonlinear Schrödinger chain at  $(T_L, T_R, \gamma) = (10, 2, 0.1)$ . Direct sampling at  $n = 10, 20, 30, 40$  resolves finite-time large-deviation scaling but loses the negative entropy tail before the asymptotic symmetry is reached. At  $n = 2$ , two independent finite-time total-entropy constructions pass detailed and integral fluctuation-relation audits. For the long-chain question, controlled cloning estimates satisfy every prospectively specified support, observation-time, population, timestep, and selection-interval gate. The resolved SCGF pairs are numerically consistent with  $\psi_n(k) = \psi_n(1 - k)$  at all four tested finite lengths. This is a scoped numerical verification, not an all-tilt proof or an infinite-chain result.

## 1 Questions and observables

The mentor’s direct-sampling question concerns both tails of a finite-time current, their threshold dependence, and their evolution with the averaging time. The thermodynamic fluctuation theorem concerns entropy production. These are related but not identical questions. In particular, the action current is not substituted for entropy production: the joint simulations show that action and bath heat are not tightly coupled at long chain lengths.

The projection-free Cartesian dynamics use  $c_j = x_j + iy_j$ . Bath heat is accumulated from the bath increment, with heat positive into the system, and the medium entropy is

$$\Sigma^{\text{m}} = -\frac{Q_L}{T_L} - \frac{Q_R}{T_R}. \quad (1)$$

For the stationary long-time test we use the right-bath entropy-current gauge  $\Sigma_R = -(T_R^{-1} - T_L^{-1})Q_R = -0.4Q_R$ . Its SCGF should obey the Gallavotti–Cohen relation

$$\psi_n(k) = \psi_n(1 - k). \quad (2)$$

## 2 Direct two-tail experiment

For each  $n = 10, 20, 30, 40$ , the production data contain 1,000,064 non-overlapping base blocks of length 20, aggregated without overlap to  $t = 20, 40, \dots, 200$ . Every block stores both bath heats,

the energy change, medium entropy, action current, stream and block identifiers, and the first-law residual. An independent recomputation performed 75,617 checks with zero errors.

At fixed threshold, every raw-count-qualified action-current tail fit obeys  $\log P_t \simeq b - tI(A)$  with  $R^2 \geq 0.98$ . Thus the data give clear finite-time large-deviation evidence. The centre becomes more Gaussian-looking with time, but the  $t = 20$  far tails are heavier than the Gaussian fixed by the full-sample variance. Direct medium-entropy and heat symmetry slopes move toward, but do not reach, their asymptotic reference before negative-tail counts disappear. The direct conclusion is therefore “not verified at accessible times,” not a violation.

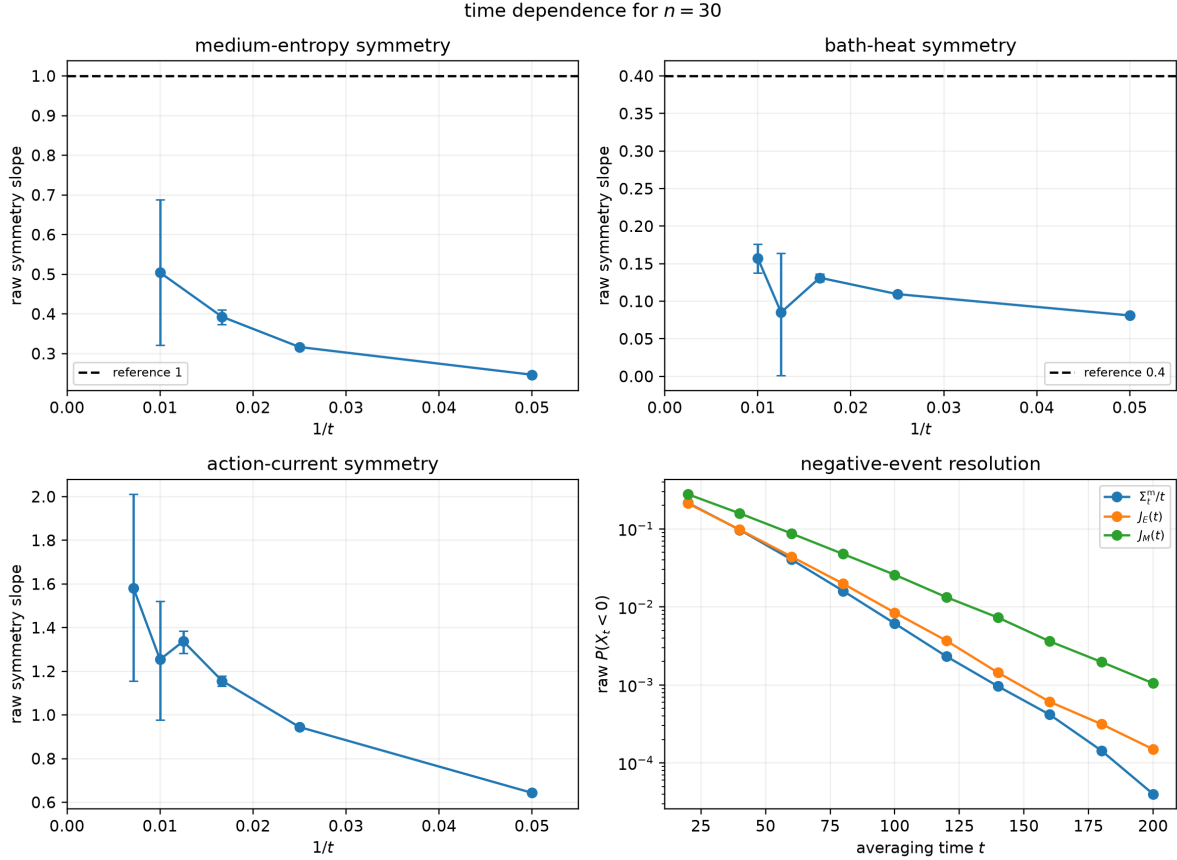


Figure 1: Representative  $n = 30$  finite-time diagnostics from the mentor’s direct-sampling experiment. Probability-ratio fits are restricted to bins with resolved counts; smoothed or Gaussian-extrapolated tails are not used to claim a fluctuation theorem.

### 3 Finite-time total entropy for the two-site chain

The finite-time total entropy adds the stationary endpoint contribution,  $\Delta s_{\text{tot}} = \Sigma^m + \Delta s_{\text{sys}}$ . The accepted parametric NESS endpoint analysis uses 1,048,576 blocks per condition. Its detailed-FT slope is  $0.994496 \pm 0.00738$ ; the driven integral-FT estimate is  $\log \langle e^{-\Delta s_{\text{tot}}} \rangle = 0.002175$  with 95% interval  $[-0.000744, 0.005251]$ . Equilibrium density validation, model sensitivity, stationarity, support, and numerical-integrity gates all pass.

An independent discrete path-ratio calculation gives a Crooks slope  $0.993352 \pm 0.00243$ . Its forward and reverse integral relations pass, and the kernel entropy approaches the physical heat entropy at first order in the timestep. The agreement of the endpoint and path-ratio constructions is a strong finite-time small-chain control; it does not establish the long-chain limit.

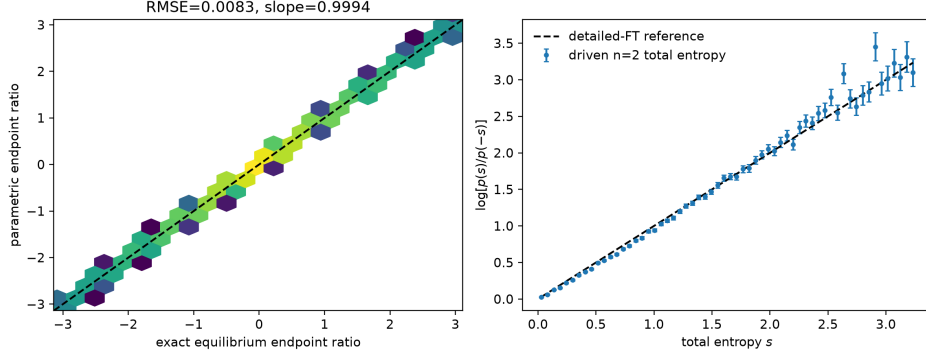


Figure 2: Accepted parametric  $n = 2$  NESS total-entropy symmetry analysis.

## 4 Controlled long-chain SCGF test

The rare-event study fixes four independent seeds for each tilt member and requires support, full-window and late-half symmetry, observation-time convergence, and the relevant population, timestep, and selection-interval controls. Phase II supplied the missing long-chain controls without changing the frozen tilts or tolerances. All 56 Phase-II summaries, timeseries, and timing logs are complete and finite, with zero midpoint failures.

| $n$ | $(k, 1 - k)$ | $N_c$ | $t$ | $D_n(k)$  | 95% CI                | controls |
|-----|--------------|-------|-----|-----------|-----------------------|----------|
| 10  | (0.3, 0.7)   | 2048  | 60  | -0.000075 | [-0.009953, 0.009802] | pass     |
| 20  | (0.4, 0.6)   | 1024  | 60  | -0.003389 | [-0.012081, 0.005302] | pass     |
| 30  | (0.4, 0.6)   | 4096  | 60  | -0.002336 | [-0.007251, 0.002578] | pass     |
| 40  | (0.4, 0.6)   | 1024  | 120 | +0.004498 | [-0.002343, 0.011340] | pass     |

Table 1: Final paired residuals  $D_n(k) = \psi_n(k) - \psi_n(1 - k)$ . “Pass” means that the complete predeclared support, time, population, timestep, and selection suite applicable to that row passes.

At  $n = 30$ , increasing the population from  $N_c = 2048$  to 4096 passes both individual-member gates and the paired-residual gate. At the accepted  $N_c = 4096$  baseline, timestep halving changes the paired residual by  $-0.001902$ , while reducing the selection interval changes it by  $-0.003157$ ; both changes pass the absolute and two-standard-error criteria. All four individual control comparisons also pass.

The normalized residual magnitudes are approximately 0.04%, 4.0%, 4.9%, and 12.9% for  $n = 10, 20, 30, 40$ . The  $n = 40$  absolute gate passes, but the relative uncertainty is larger because the SCGF itself is small. This is an important effect-size caveat rather than evidence of a resolved violation.

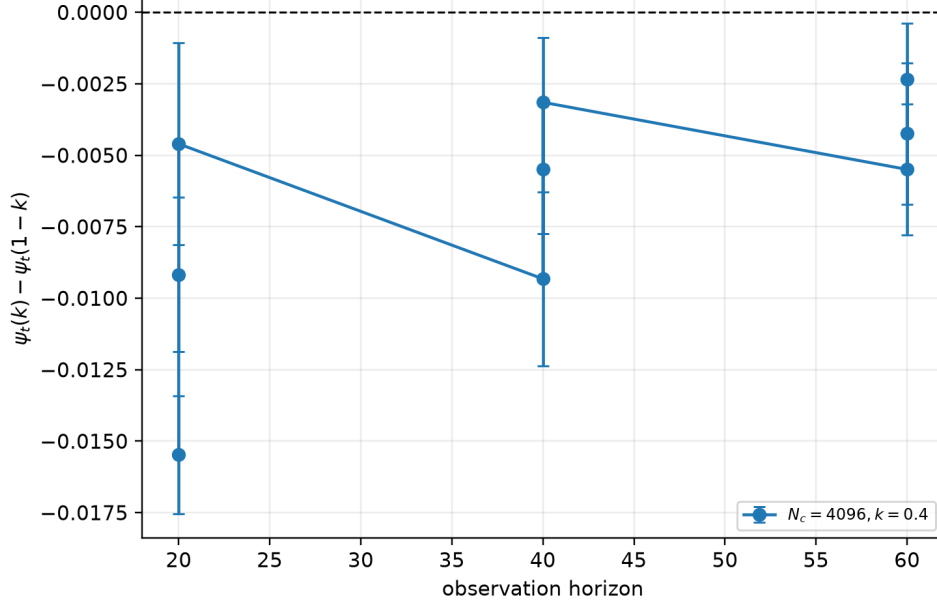


Figure 3: The  $n = 30$  paired SCGF residual across the tested observation horizons and numerical configurations. The final population, timestep, and selection controls all retain support and pass the frozen criteria.

## 5 Interpretation and allowed claim

Direct sampling and cloning answer complementary parts of the same problem. The former establishes finite-time large-deviation structure but becomes rare-tail limited. The latter resolves a preselected asymptotic SCGF pair and passes the required numerical controls. Together they support the statement:

For the boundary-driven projection-free Cartesian NLS chain at  $(T_L, T_R, \gamma) = (10, 2, 0.1)$ , controlled rare-event estimates are consistent with Gallavotti–Cohen SCGF symmetry at all four tested finite chain lengths over the resolved complementary tilt pair.

The result does not prove exact equality, cover every  $k \in [0, 1]$ , establish uniform accuracy as  $n \rightarrow \infty$ , or identify action current with entropy production. A stronger numerical claim would require a separately frozen grid of additional tilts; an infinite-chain claim would additionally require a dedicated finite-size study of the symmetry residual.

## Reproducibility

The prospective protocol, run matrix, raw Phase-II outputs, generated analysis tables, analysis-grid erratum, material passport, and final verdict are archived beside this report. The simulator and analysis-engine hashes are recorded in the validation report. The source self-tests passed before production, and no failed cell was followed by data-dependent retuning.